Data Structures & Algorithms (CST-003) - Unit 2

Unit 2 - Stacks & Queues - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

What this unit covers: stacks and queues - the two restricted-access lists, with their applications.

Key ideas: A **stack** is LIFO: push and pop happen at the same end (top). Applications: function-call stack, undo, bracket matching, and expression conversion. Learn **infix to postfix** by hand: operands go straight to output; operators go on the stack by precedence (pop higher-or-equal precedence first); '(' pushes, ')' pops until '('. Then **evaluate postfix**: push operands, on an operator pop two, apply, push back. A **queue** is FIFO: insert at rear, delete at front. A **circular queue** wraps the array around to reuse freed slots - know the (index + 1) % size formula and the full/empty conditions. Variants: **deque** (insert/delete at both ends) and **priority queue** (highest priority leaves first).

Exam pointers: converting and evaluating one expression by hand (showing the stack at each step) is almost guaranteed. Also common: write push/pop or enqueue/dequeue as functions, and "why does a simple queue waste space but a circular queue does not".

Made by Anas. Free forever. Watch the linked videos on the site after reading.